$$\text{Ratio of roots} = \frac{-1+\sqrt{3}}{-1-\sqrt{3}} = \frac{\omega}{\omega^2} = \frac{1}{\omega}$$

$$\text{Product of roots} = \frac{(-1+\sqrt{3})}{2}\frac{(-1-\sqrt{3})}{2}\times 4$$

$$= \omega.\omega^2.4$$

$$= 4 = \frac{b^2}{a^2}$$

33. Option (c) is correct.

Let n be an integer and $n = 0$

$\therefore \quad x^2 + mn = 0$

$\Rightarrow \quad x(x + m) = 0$

$\Rightarrow \quad x = 0, x = -m$

Since n is integer therefore m is also integer.

34. Option (a) is correct.

$(x + y)^{2n+1}.(x - y)^{2n+1} = (x^2 - y^2)^{2n+1}$

Middle term

$$\frac{2n+2}{2}, \frac{2n+4}{2} = (n+1)^{\text{th term}}, (n+2)^{\text{th term}}$$

$$T_{n+1} = {}^{2n+1}C_n (x^2)^{n+1}(y^2)^n$$

$$T_{n+2} = {}^{2n+1}C_{n+1}(x^2)^n(y^2)^{n+1}$$

$${}^{2n+1}C_n(x^2)^{n+1}(y^2)^n = {}^{2n+1}C_{n+1}(x^2)^n(y^2)^{n+1}$$

$$\Rightarrow \frac{{}^{2n+1}C_n}{{}^{2n+1}C_{n+1}} = \frac{x^{2n}y^{2n+2}}{x^{2n+2}y^{2n}}$$

$$\Rightarrow \frac{(2n+1)!}{(n+1)!n!}\times\frac{n!(n+1)!}{(2n+1)!} = \frac{y^2}{x^2}$$

$$\Rightarrow \frac{y^2}{x^2} = 1$$

35. Option (c) is correct.

$n(A) = 5$ and $n(B) = 2$

Number of onto functions $= 2^5 - {}^2C_1(2-1)^5$

$= 32 - 2 = 30$

36. Option (d) is correct.

$$\frac{\sqrt{3}\cos 10^\circ - \sin 10^\circ}{\sin 25^\circ.\cos 25^\circ}$$

$$= \frac{2\left(\frac{\sqrt{3}}{2}.\cos 10^\circ - \frac{1}{2}\sin 10^\circ\right)}{\frac{1}{2}(2\sin 25^\circ.\cos 25^\circ)}$$

$$= \frac{4.\sin(60^\circ - 10^\circ)}{\sin 50^\circ}$$

$$= \frac{4\sin 50^\circ}{\sin 50^\circ}$$

$$= 4$$

37. Option (c) is correct.

$$\sin 9^\circ - \cos 9^\circ = \sqrt{2}\left(\frac{1}{\sqrt{2}}\sin 9^\circ - \frac{1}{\sqrt{2}}\cos 9^\circ\right)$$

$$= \sqrt{2}\sin(45^\circ - 9^\circ)$$

$$= \sqrt{2}\sin 36^\circ$$

$$= \sqrt{2}\sqrt{1-\cos^2 36^\circ}$$

$$= \sqrt{2}\sqrt{1-\left(\frac{\sqrt{5}+1}{4}\right)^2}$$

$$\left[\because \cos 36^\circ = \frac{\sqrt{5}+1}{4}\right]$$

$$= \sqrt{2}\frac{\sqrt{10-2\sqrt{5}}}{4}$$

$$= \frac{\sqrt{5-\sqrt{5}}}{2}$$

38. Option (d) is correct.

Given that,

$\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A . \sin B . \sin C$

$\Rightarrow \sin A + \sin B + \sin C = 0$

$\Rightarrow \quad ak + kb + kc = 0$

$\Rightarrow \quad a + b + c = 0$

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a+b+c & b & c \\ b+c+a & c & a \\ c+a+b & a & b \end{vmatrix}$$

(Applying $C_1 \to C_1 + C_2 + C_3$)

$$= \begin{vmatrix} 0 & b & c \\ 0 & c & a \\ 0 & a & b \end{vmatrix}$$

$$= 0$$

39. Option (c) is correct.

$$\cos^{-1}x = \sin^{-1}x$$

$$\Rightarrow \frac{\pi}{2} - \sin^{-1}x = \sin^{-1}x$$

$$\Rightarrow 2\sin^{-1}x = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}x = \frac{\pi}{4}$$

$$\Rightarrow x = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

40. Option (b) is correct.

$(\sin\theta - \cos\theta)^2 = 2$

$\Rightarrow \sin^2\theta + \cos^2\theta - 2\sin\theta.\cos\theta = 2$

$\Rightarrow \quad \sin 2\theta = -1$

$\therefore$ Two solutions.